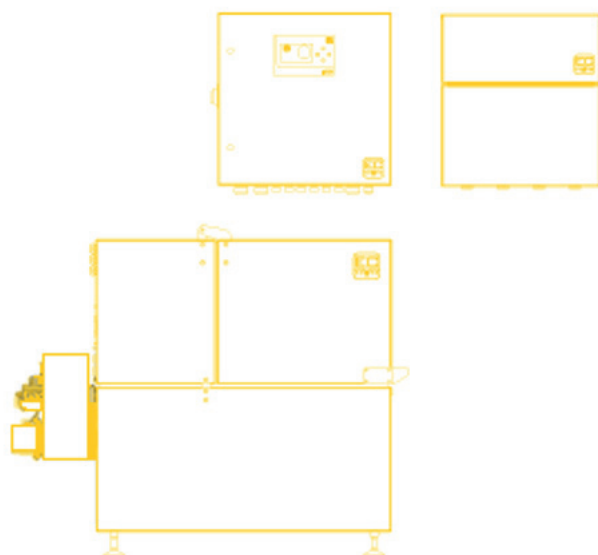
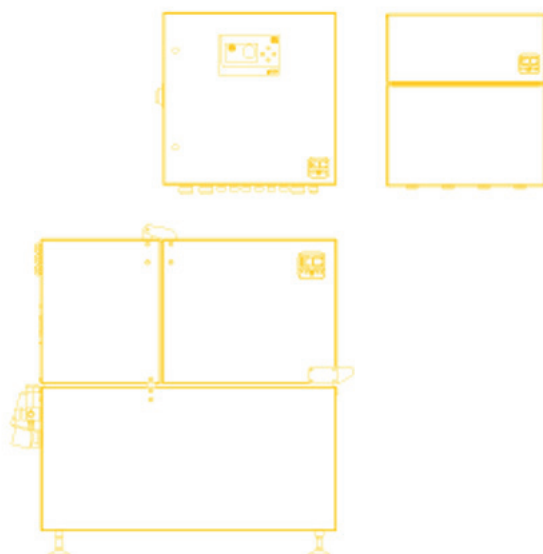




XRGI[®] 25

TECHNICAL DATA

Q70 iQ25



The XRGi® 25 is a heating solution for space heat and domestic hot water designed to generate electricity while producing heat. This concept of simultaneously producing heat and power is known as Combined Heat and Power (CHP).

The main component of the system is a highly efficient combustion engine which generates electricity while capturing and transferring the heat from its cooling system and exhaust gas to the heating system of the building.

The XRGi® 25 operates on natural gas or propane and consists of three main components:

- Power Unit
- Q-Heat Distributor
- iQ25-Control Panel

The XRGi® 25 will provide the required heat for the building and generate significant savings on electric bills as well as reduce the carbon footprint through onsite power generation.

ORDERING DATA

Manufacturer	EC POWER
Model	XRGi® 25
Article number	X250001
Modules	Power Unit, iQ25-Control Panel, Q70-Heat Distributor

OUTPUT

Power Modulation		50%	75%	100%
Nominal electricity capacity, modulating	kW	12	18	24
Parasitic power	kW	0.08	0.08	0.08
Net electricity capacity, W_E	kW	11.92	17.92	23.92
Thermal output, Q_{TH}	kW	33	38	48
	MBH	112.6	129.7	163.8
Fuel energy input, Q_{FUEL} (LHV/HHV) ¹	kW	48 / 53	60 / 67	77 / 86
	MBH	163.8 / 182.0	204.7 / 227.5	262.7 / 291.9

EFFICIENCIES

Power Modulation		50%	75%	100%
Electrical efficiency (LHV/HHV)	%	25.0 / 22.5	30.0 / 27.0	31.2 / 28.1
Net electrical efficiency (LHV/HHV)	%	24.8 / 22.4	29.9 / 26.9	31.1 / 28.0
Thermal efficiency (LHV/HHV)	%	68.8 / 61.9	63.3 / 57.0	62.3 / 56.1
Total CHP efficiency (LHV/HHV) ²	%	93.6 / 84.2	93.2 / 83.9	93.4 / 84.1
Effective electrical efficiency (LHV/HHV) ³	%	109.6 / 98.6	103.9 / 93.5	104.0 / 93.6
Electrical heat rate (LHV/HHV) ⁴	Btu/kWh	13,740 / 15,266	11,424 / 12,693	10,983 / 12,204
Net heat rate (LHV/HHV) ⁵	Btu/kWh	3,113 / 3,459	3,284 / 3,649	3,281 / 3,645
Power/heat ratio		0.36	0.47	0.50

1 LHV (lower heating value/HHV (higher heating value) ratio: 0.9
Btu/kWh: 3,412

2 Total CHP efficiency is the sum of the net useful power output and net useful thermal output divided by the total fuel input.

3 Effective electric efficiency measures how effectively the CHP system generates power once the thermal demand of a site has been met and hence allow for a direct comparison of CHP to conventional power generation system performance:

$$\mathcal{E}_{EE} = \frac{W_E}{Q_{FUEL} - \sum (Q_{TH} / \alpha)} \quad , \alpha \text{ (HHV)} = 0.8$$

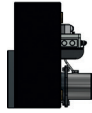
α equals the efficiency of a gas boiler which otherwise would be used to produce the useful thermal energy output if the CHP system did not exist.

4 Electrical heat rate is the amount of energy used by a CHP to generate one kilowatt-hour (kWh) of electricity.

5 Net heat rate is the incremental electric generating efficiency of a CHP and is the heat rate normally used to calculate the cost of fuel for electricity generation:

$$\text{Net heat rate} = \frac{Q_{FUEL} - \sum (Q_{TH} / \alpha)}{W_E}$$

Q70 iQ25



BW8+

Upgrade your XRGi® system with BW8+ condensing technology and experience a new level of efficiency and performance.

Capture and utilize more heat from exhaust gases, significantly boosting overall system efficiency.

Lower your carbon footprint with reduced CO₂ and NO_x emissions per energy output, contributing to a cleaner environment.

ORDERING DATA

Manufacturer	EC POWER
Model	XRGi® 25 & BW8+
Article number	X250001 & K000105
Modules	Power Unit, iQ25-Control Panel, Q70-Heat Distributor, BW8+

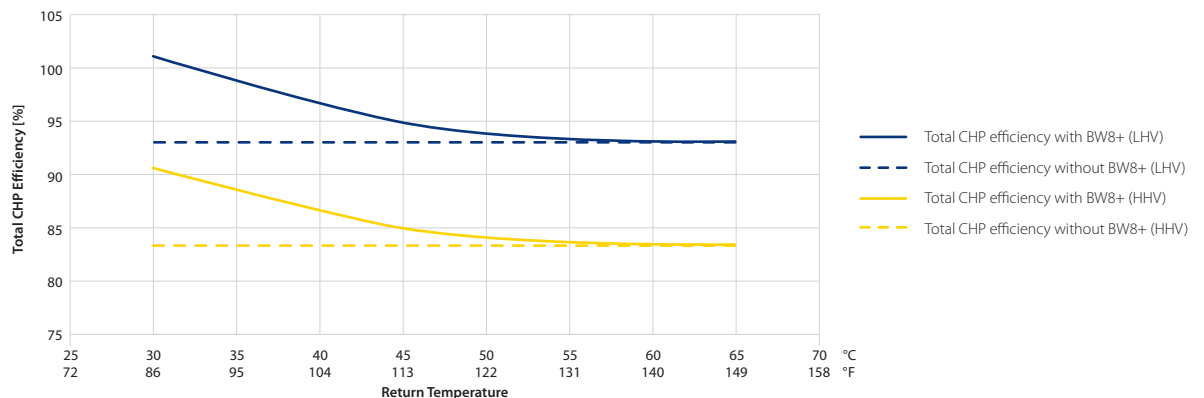
OUTPUT

Power Modulation		50%	75%	100%
Nominal electricity capacity, modulating	kW	12	18	24
Parasitic power	kW	0.08	0.08	0.08
Net electricity capacity, W_E	kW	11.92	17.92	23.92
Thermal output, Q_{TH} at 30 °C / 86 °F return temperature	kW	36	43	54
	MBH	123.5	145.0	184.2
Fuel energy input, Q_{FUEL} (LHV/HHV) ¹	kW	48 / 53	60 / 67	77 / 86
	MBH	163.8 / 182.0	204.7 / 227.5	262.7 / 291.9

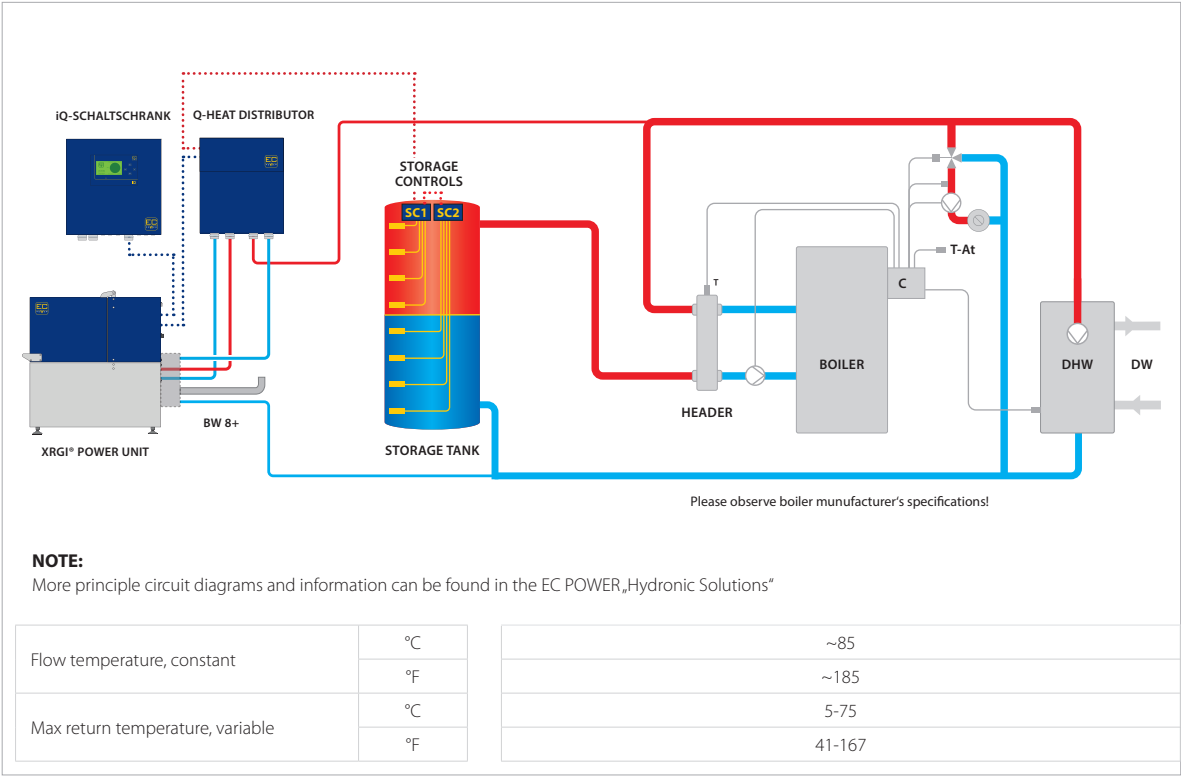
EFFICIENCIES

Power Modulation		50%	75%	100%
Electrical efficiency (LHV/HHV)	%	25.0 / 22.5	30.0 / 27.0	31.2 / 28.1
Net electrical efficiency (LHV/HHV)	%	24.8 / 22.4	29.9 / 26.9	31.1 / 28.0
Thermal efficiency (LHV/HHV)	%	75.4 / 67.9	70.8 / 63.8	70.1 / 63.1
Total CHP efficiency (LHV/HHV) ²	%	100.3 / 90.2	100.7 / 90.6	101.2 / 91.1
Effective electrical efficiency (LHV/HHV) ³	%	163.8 / 147.5	147.0 / 132.3	147.2 / 132.5
Electrical heat rate (LHV/HHV) ⁴	Btu/kWh	13,740 / 15,266	11,424 / 12,693	10,983 / 12,204
Net heat rate (LHV/HHV) ⁵	Btu/kWh	2,082 / 2,314	2,321 / 2,578	2,318 / 2,575
Power/heat ratio		0.33	0.42	0.44

XRGi® 25 TOTAL CHP EFFICIENCY AT FULL LOAD



HYDRONIC
INTEGRATION



FUELS

Natural gas (all qualities), propane, butane
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EXHAUST GAS

Max exhaust gas temperature	°C	120
	°F	248
Emissions, NOx	kg/MWh _{e+t}	<0.03
	lbs/MWh _{e+t}	<0.06 (CARB DG Reg. 0.07)
Emissions, CO	kg/MWh _{e+t}	<0.03
	lbs/MWh _{e+t}	<0.06 (CARB DG Reg. 0.10)
Emissions, VOC	kg/MWh _{e+t}	<0.003
	lbs/MWh _{e+t}	<0.006 (CARB DG Reg. 0.02)

POWER CONNECTION

Voltage, 3 phase AC	V AC	480
Frequency	Hz	60

SERVICE

Service interval (operating hours)	4,000
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DIMENSIONS
AND WEIGHT

		XRG1® 25 Power Unit	Q70-Heat Distributor	iQ25-Control Panel
Dimensions, W x H x D	mm	750 x 1,170 x 1,120	550 x 600 x 295	600 x 600 x 210
	in	29.5 x 44.1 x 46.1	21.6 x 23.6 x 11.6	23.6 x 23.6 x 8.3
Footprint	m²	0.84	wall mounted	wall mounted
	sqft	9		
Weight	kg	750	44	40
	lbs	1,500	97	88



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XRGI[®] 25

TECHNICAL DATA